Evaluation of increment PDFs using Laplace's method

Abstract

We derive asymptotic approximations for velocity increment PDFs in the multifractal propagator framework. Using Laplace's method, we obtain two solutions: a simplified form assuming a fixed saddle point, and a better approximation via the Lambert W function. Numerical tests against the full propagator integral reveal where each approximation works best, providing groundwork for entropy calculations.

1 Introduction

Turbulent flows exhibit intermittency: velocity fluctuations become increasingly non-Gaussian at smaller scales. The fluctuations at scale ℓ can be studied with the velocity increments equation.

The multifractal approach describes intermittency through local Hölder exponents h distributed according to a spectrum $D_E(h)$. Chevillard et al. [1] proposed a propagator formalism where increment PDFs appear as integrals over the Hölder distribution, allowing for Gaussian large-scale statistics with small-scale intermittency.

We use the Kolmogorov–Oboukhov lognormal model with $D_E(h) = 1 - (h - c_1)^2/(2c_2)$. This gives tractable expressions while capturing intermittency. Since the propagator integrals lack closed forms, we apply Laplace's method for asymptotic approximations at small scales.

The paper proceeds as follows. Section 2 establishes the Hölder distribution and propagator integral. Section 3 derives two approximations of the increment PDFs: assuming a fixed saddle point (Case A), and then using the Lambert W function (Case B). Section 4 introduces standardized PDFs. Section 5 addresses entropy. We compare both approximations numerically against the full integral to assess their accuracy.

2 Starting point: the distribution of h

The multifractal formalism for velocity increments can be interpreted as a continuous mixture of Gaussian variables. More precisely, the velocity increment at scale ℓ, denoted as $\delta_\ell u$, is modeled as a Gaussian random variable conditioned on a fluctuating singularity exponent h (Chevillard et al., 2012, Eq. 10) :

$$\delta_\ell u \overset{\text{law}}{=} \sigma \left(\frac{\ell}{L}\right)^h \delta \tag{0}$$

where σ represents the characteristic velocity of the flow at the integral scale L, δ is a zero-mean, unit-variance Gaussian white noise, and h is a fluctuating variable independent of δ, whose probability density function $\mathcal{P}_h^{(\ell)}$ is governed by a singularity spectrum $\mathcal{D}(h)$.

We now use (Chevillard et al., 2012, Eq. 11), which provides the probability density function of h :

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)}}{\displaystyle\int_{h_{\min}}^{h_{\max}} \left(\frac{\ell}{L}\right)^{1-\mathcal{D}^E(h)} \mathrm{d}h}. \tag{1}$$

using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1 - \frac{(h - c_1)^2}{2c_2},$$

with $h_{\min} = -\infty$ and $h_{\max} = +\infty$, we obtain

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{(h-c_1)^2}{2c_2}\ln\left(\frac{L}{\ell}\right)\right]}{\displaystyle\int_{-\infty}^{+\infty}\exp\left[-\frac{(h-c_1)^2}{2c_2}\ln\left(\frac{L}{\ell}\right)\right]\mathrm{d}h}. \tag{2}$$

To simplify notation, we define

$$\alpha = -\ln\left(\frac{\ell}{L}\right) > 0. \tag{3}$$

Then the expression becomes

$$\mathcal{P}_h^{(\ell)}(h) = \frac{\exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right]}{\displaystyle\int_{-\infty}^{+\infty}\exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right]\mathrm{d}h}. \tag{4}$$

The denominator is a Gaussian integral:

$$\int_{-\infty}^{+\infty} e^{-\frac{\alpha}{2c_2}(h-c_1)^2}\mathrm{d}h = \sqrt{\frac{2\pi c_2}{\alpha}}. \tag{5}$$

Hence,

$$\boxed{\mathcal{P}_h^{(\ell)}(h) = \sqrt{\frac{\alpha}{2\pi c_2}}\,\exp\left[-\frac{\alpha}{2c_2}(h-c_1)^2\right].} \tag{6}$$

3 Computation of the PDF using Laplace's method

3.1 General form of the PDF

We now consider the probability density function of velocity increments. It's a function of the position x. According to (Chevillard et al., 2012, Eq. 17) :

$$\mathcal{P}_{\delta_\ell u}(x) = \frac{1}{2\pi\sigma}\sqrt{\frac{\alpha}{c_2}}\int_{-\infty}^{+\infty}\exp[-f(h,\alpha,x)]\,\mathrm{d}h. \tag{7}$$

with

$$f(h,\alpha,x) = \alpha\left(\frac{(h-c_1)^2}{2c_2} - h\right) + \frac{1}{2}\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{8}$$

We would like to get an explicit form of our Eq. 7. To do so, we'll use Laplace's approximation (see Appendix). Knowing that

$$\forall h \in \mathbf{R}, \quad \frac{\partial f}{\partial h}(h,\alpha,x) = \alpha\left(\frac{h-c_1}{c_2} - 1\right) + \alpha\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{9}$$

we have :

$$\forall h \in \mathbf{R}, \quad \frac{\partial^2 f}{\partial h^2}(h,\alpha,x) = \frac{\alpha}{c_2} + 2\alpha^2\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \quad > 0 \tag{10}$$

We just need to verify that $h^\star$ is such that $\frac{\partial f}{\partial h}(h^\star,\alpha,x) = 0$ to be able to use Laplace approximation.

In this case, by applying the approximation to Eq. 7, we get :

$$\boxed{\mathcal{P}_{\delta_\ell u}(x) \approx \sqrt{\frac{\alpha}{2\pi c_2 \sigma^2 \frac{\partial^2 f}{\partial h^2}(h,\alpha,x))}}\,\exp[-f(h^\star,\alpha,x)].} \tag{11}$$

i.e

$$\boxed{\mathcal{P}_{\delta_\ell u}(x) \approx \frac{1}{\sqrt{2\pi}\sqrt{2\alpha c_2 x^2 e^{2\alpha h^\star} + \sigma^2}} \times \exp\left[-\alpha\left(\frac{(h^\star - c_1)^2}{2c_2} - h^\star\right) - \frac{1}{2}\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h^\star}\right]} \tag{12}$$

Here is a plot of both the obtained PDF and its log using Julia programming language.

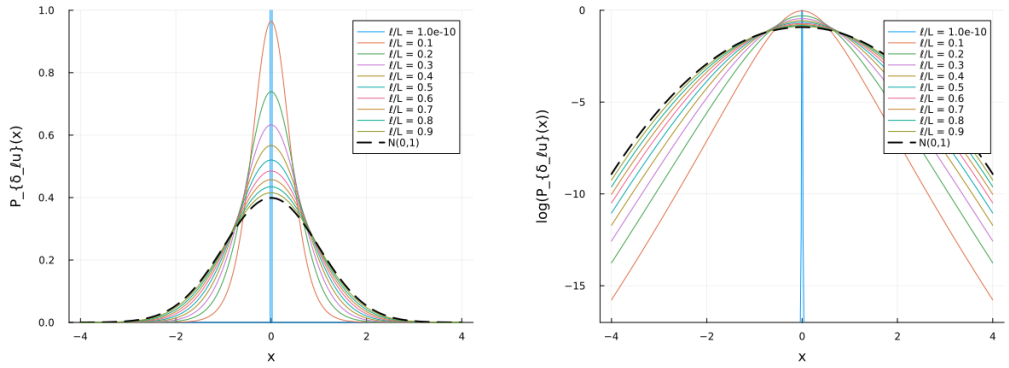


Figure 1: $\mathcal{P}_{\delta_\ell u}$ and $\log(\mathcal{P}_{\delta_\ell u})$ plots using the explicit form of the PDF (Julia, QuadGK)

In order to visualize the properties of $\mathcal{P}_{\delta_\ell u}$, we plot its second order structure, flatness and skewness.

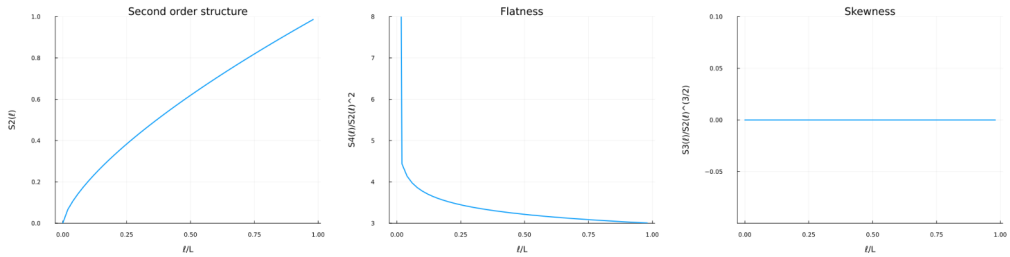


Figure 2: Second order structure, flatness and skewness plots using the explicit form of $\mathcal{P}_{\delta_\ell u}$ (Julia, QuadGK)

3.2 Case A: assuming $h^\star = c_1$

As a first approximation, we can consider that :

$$\frac{\partial f}{\partial h}(h^\star, \alpha, x)\Big|_{h^\star = c_1} = 0 \tag{13}$$

We can use the Laplace approximation given by Eq. 12, which becomes :

$$\boxed{\mathcal{P}_{\delta_\ell u}(x) \approx \frac{1}{\sqrt{2\pi}} \frac{1}{\sqrt{2\alpha c_2 x^2 e^{2\alpha c_1} + \sigma^2}} \times \exp\left[\alpha c_1 - \frac{1}{2}\left(\frac{x}{\sigma}\right)^2 e^{2\alpha c_1}\right]} \tag{14}$$

We plot again both the obtained PDF and its log using Julia programming language. The approximation seems to be quite accurate since the slopes are very similar to the previous plots.

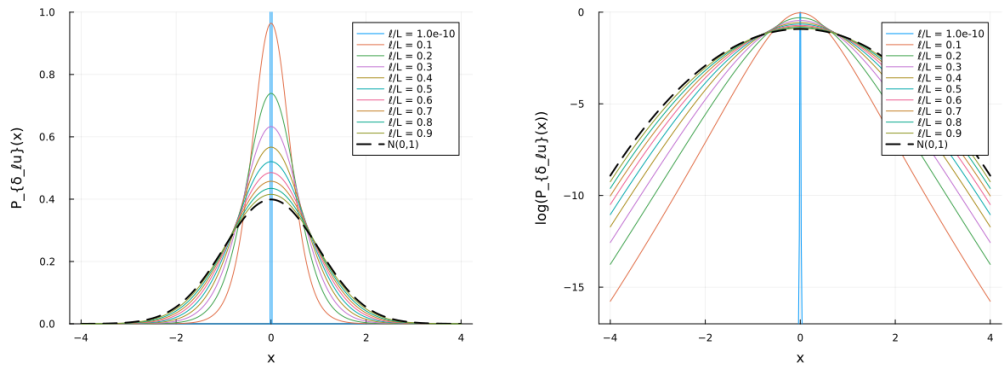


Figure 3: $P_{\delta_\ell u}$ and $\log(P_{\delta_\ell u})$ plots using Laplace approximation of the PDF with $h^* = c_1$ (Julia, QuadGK)

3.3 Case B: solving $\frac{\partial f}{\partial h}(h, \alpha, x) = 0$ via the Lambert W function

From (8),

$$\frac{\partial f}{\partial h}(h, \alpha, x) = \alpha\left(\frac{h - c_1}{c_2} - 1\right) + \alpha\left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{15}$$

Set to zero:

$$h - (c_1 + c_2) = -c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha h} \tag{16}$$

Multiply by -2α and let $u = -2\alpha(h - c_1 - c_2)$:

$$u = 2\alpha c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)-u} \tag{17}$$

Consequently

$$ue^u = 2\alpha c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)} \tag{18}$$

Hence

$$\boxed{h^\star(\alpha, x) = (c_1 + c_2) - \frac{1}{2\alpha} W\left(2\alpha c_2 \left(\frac{x}{\sigma}\right)^2 e^{2\alpha(c_1+c_2)}\right)} \tag{19}$$

where $W(x)$ is the Lambert W function (the inverse of $w \mapsto we^w$).

We can now plug the obtained $h^\star$ into Eq. 12. We get these plots using Julia programming language.

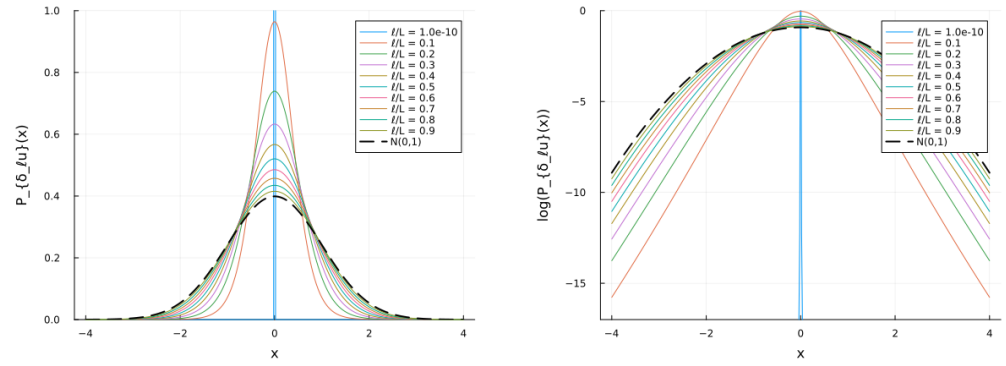


Figure 4: $P_{\delta_\ell u}$ and $\log(P_{\delta_\ell u})$ plots using Laplace approximation with h^* from Lambert W (Julia, QuadGK)

4 Normalized PDF

According to (Chevillard et al., 2012, Eq. 12), the p-th order moment of the velocity increment is expressed as

$$\langle|\delta_\ell u|^p\rangle \underset{\ell\to 0}{\approx} \sigma^2\langle|\delta|^p\rangle \left(\frac{\ell}{L}\right)^{\zeta_p}, \quad \text{with} \quad \zeta_p = \min_h\big[ph+1-\mathcal{D}^E(h)\big] \quad \text{and} \quad \delta\sim\mathcal{N}(0,1). \tag{20}$$

Using the Kolmogorov–Oboukhov log-normal spectrum

$$\mathcal{D}^E(h) = 1-\frac{(h-c_1)^2}{2c_2}, \tag{21}$$

the minimization problem is

$$\frac{\mathrm{d}}{\mathrm{d}h}\big[ph+1-\mathcal{D}^E(h)\big]=0 \quad i.e. \quad \frac{\mathrm{d}}{\mathrm{d}h}\big[ph+\frac{(h-c_1)^2}{2c_2}\big]=0 \tag{22}$$

Hence, the minimizing exponent is

$$h_p = c_1 - pc_2, \tag{23}$$

and inserting this value into Eq. 20 gives

$$\zeta_p = pc_1 - \frac{p^2}{2}c_2 \tag{24}$$

In particular, for the second-order :

$$\zeta_2 = 2(c_1-c_2) \tag{25}$$

Knowing that $\langle|\delta|^2\rangle = 1$, we get :

$$\boxed{\langle(\delta_\ell u)^2\rangle \underset{\ell\to 0}{\approx} \sigma^2\left(\frac{\ell}{L}\right)^{2(c_1-c_2)}}. \tag{26}$$

We now define a normalized increment

$$\boxed{\delta_\ell\nu = \frac{\delta_\ell u}{\sqrt{\langle(\delta_\ell u)^2\rangle}} = \frac{\delta_\ell u}{\sigma\left(\frac{\ell}{L}\right)^{c_1-c_2}}} \tag{27}$$

This normalization ensures that

$$\langle(\delta_\ell\nu)^2\rangle = 1, \tag{28}$$

for all scales ℓ/L, so that all normalized PDFs can be directly compared across scales.

We thus have $\delta_\ell\nu = \delta_\ell u/\sigma\left(\frac{\ell}{L}\right)^{c_1-c_2} = g(\delta_\ell u)$ where $g(x) = x/\sigma\left(\frac{\ell}{L}\right)^{c_1-c_2}$ is monoton and differentiable.

Knowing that : $\mathcal{P}_{\delta_\ell\nu}(x) = \left|\frac{1}{g'(g^{-1}(x))}\right|\mathcal{P}_{\delta_\ell u}(g^{-1}(x))$, we have : $\mathcal{P}_{\delta_\ell\nu}(x) = \sigma\left(\frac{\ell}{L}\right)^{c_1-c_2}\mathcal{P}_{\delta_\ell u}(\sigma\left(\frac{\ell}{L}\right)^{c_1-c_2}x)$ i.e.

$$\boxed{\mathcal{P}_{\delta_\ell\nu}(x) = \sigma e^{\alpha(c_2-c_1)}\mathcal{P}_{\delta_\ell u}(\sigma e^{\alpha(c_2-c_1)}x)} \tag{29}$$

We then obtain from Eq. 12 the normalized form of the PDF :

$$\boxed{\mathcal{P}_{\delta_\ell\nu}(x) \approx \frac{e^{\alpha(c_2-c_1)}}{\sqrt{2\pi}\sqrt{2\alpha c_2 x^2 e^{2\alpha(h^\star+c_2-c_1)}+1}} \times \exp\left[-\alpha\left(\frac{(h^\star-c_1)^2}{2c_2}-h^\star\right)-\frac{1}{2}x^2e^{2\alpha(h^\star+c_2-c_1)}\right]} \tag{30}$$

which is valid if $h^\star$ is such that $\frac{\partial f}{\partial h}(h^\star,\alpha,\sigma e^{\alpha(c_2-c_1)}x)=0$.

As in the non-normalized case, here are the plots of both the obtained PDF and and its log using Julia programming language.

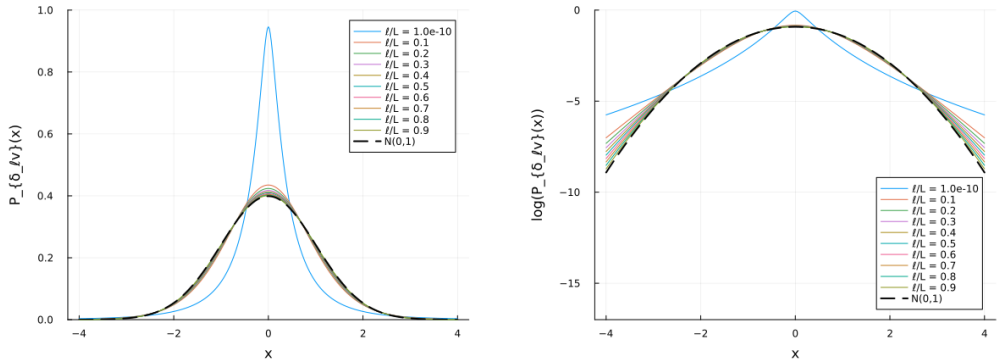


Figure 5: $P_{\delta_\ell \nu}$ and $\log(P_{\delta_\ell \nu})$ plots using the explicit form of the PDF (Julia, QuadGK)

We provide the second order structure, flatness and skewness plots of $\mathcal{P}_{\delta_\ell \nu}$ as well.

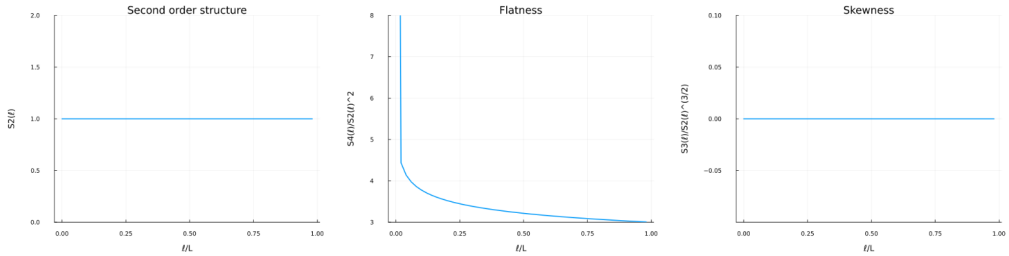


Figure 6: Second order structure, flatness and skewness plots using the explicit form of $P_{\delta_\ell \nu}$ (Julia, QuadGK)

4.1 Case A: assuming $h^\star = c_1$

If we consider that $\frac{\partial f}{\partial h}(h^\star, \alpha, \sigma e^{\alpha(c_2 - c_1)} x)\Big|_{h^\star = c_1} = 0$, then we get from Eq. 30 :

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(x) \approx \frac{1}{\sqrt{2\pi}\sqrt{2\alpha c_2\, x^2 e^{2\alpha c_2} + 1}} \times \exp\left[\alpha c_2 - \frac{1}{2} x^2 e^{2\alpha c_2}\right]} \tag{31}$$

We plot both the obtained PDF and its log using Julia programming language. The approximation seems to be quite accurate again since the slopes are very similar to the previous plots.

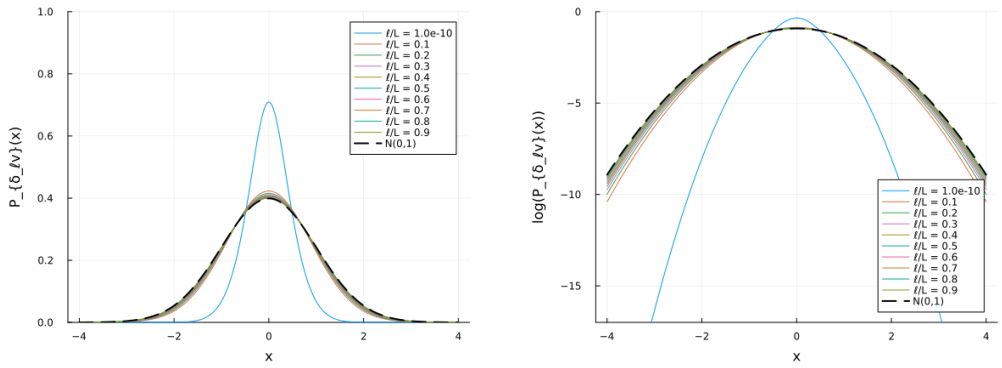


Figure 7: $P_{\delta_\ell \nu}$ and $\log(P_{\delta_\ell \nu})$ plots using Laplace approximation of the PDF with $h^* = c_1$ (Julia, QuadGK)

4.2 Case B: solving $\frac{\partial f}{\partial h}(h, \alpha, \sigma e^{\alpha(c_2-c_1)}x) = 0$ via the Lambert W function

We want to find $h^\star$ such that $\frac{\partial f}{\partial h}(h^\star, \alpha, \sigma e^{\alpha(c_2-c_1)}x) = 0$. Thanks to Eq. 19, we have :

$$\boxed{h^\star(\alpha, \sigma e^{\alpha(c_2-c_1)}x) = (c_1 + c_2) - \frac{1}{2\alpha} W\left(2\alpha c_2 x^2 e^{4\alpha c_2}\right)} \tag{32}$$

We can now plug the obtained $h^\star$ into Eq. 30. Let $W = W\left(2\alpha c_2 x^2 e^{4\alpha c_2}\right)$. We get :

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(x) \approx \frac{1}{\sqrt{2\pi}\sqrt{2\alpha c_2 x^2 e^{4\alpha c_2 - W} + 1}} \times \exp\left[-\frac{\alpha}{2}\left(\frac{1}{c_2}\left(\frac{W}{2\alpha}\right)^2 - 3c_2\right) - \frac{1}{2}x^2 e^{4\alpha c_2 - W}\right]} \tag{33}$$

i.e.

$$\boxed{\mathcal{P}_{\delta_\ell \nu}(x) \approx \frac{e^{\frac{3}{2}} c_2 \alpha}{\sqrt{2\pi}} \frac{e^{-\frac{2W+W^2}{8c_2\alpha}}}{\sqrt{1+W}}} \tag{34}$$

We get the following plots using Julia programming language.

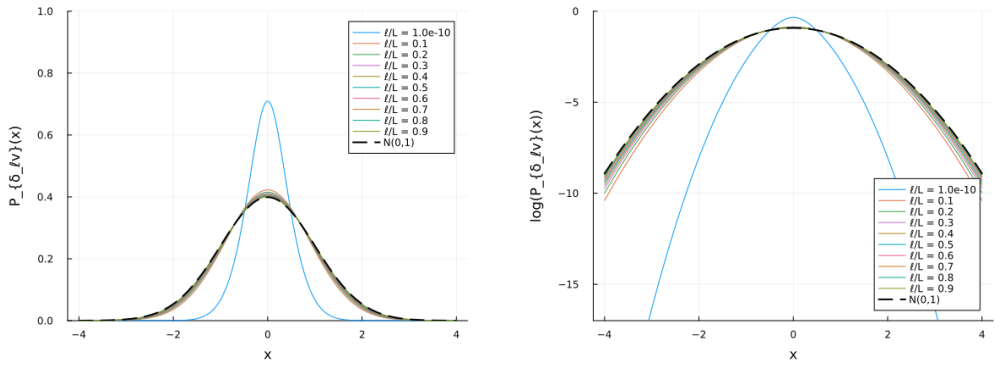


Figure 8: $P_{\delta_\ell \nu}$ and $\log(P_{\delta_\ell \nu})$ plots using Laplace approximation with h^* from Lambert W (Julia, QuadGK)

5 Entropy of the normalized PDF

We now compute the Shannon entropy of the normalized increment PDF $\mathcal{P}_{\delta_\ell \nu}$ obtained in Eq. 28. By definition,

$$S(\delta_\ell \nu) = -\int_{-\infty}^{+\infty} \mathcal{P}_{\delta_\ell \nu}(x)\, \ln\big[\mathcal{P}_{\delta_\ell \nu}(x)\big]\, \mathrm{d}x \tag{35}$$

References

[1] L. Chevillard, B. Castaing, A. Arneodo, E. Lévêque, J.-F. Pinton, and S. G. Roux, *A phenomenological theory of Eulerian and Lagrangian velocity fluctuations in turbulent flows*, C. R. Physique **13**, 899–928 (2012).

A Appendix - Laplace's approximation theorem

Let $I = [a, b]$ be an interval, $\varphi \in C^2(I, \mathbb{R})$ and $g \in C^0(I, \mathbb{C})$. Assume that:

(i) For all $t > 0$, $\displaystyle\int_a^b e^{t\varphi(x)} |g(x)|\, dx < +\infty$;

(ii) φ' vanishes at a unique point $x_0 \in I$ and $\varphi''(x_0) < 0$;

(iii) $g(x_0) \neq 0$.

Then

$$\int_a^b e^{\,t\varphi(x)}\, g(x)\, dx \sim \sqrt{\frac{2\pi}{t\,|\varphi''(x_0)|}}\, e^{\,t\varphi(x_0)}\, g(x_0) \qquad \text{as } t \to +\infty.$$